FOOD VS FUEL

THE IMPACT OF BIOFUELS ON FOOD SECURITY IN THE GLOBAL SOUTH

Jyoshil Pitamber, Managing Director, Gusheshe Fuels

The "Food vs Fuel" debate has long questioned whether growing crops for energy might compete with global food needs. It's an important concern, but evidence increasingly suggests that the real challenge lies in how much food is wasted, rather than how much is converted into biofuel. Across South Africa and the world, nearly one-third of all food produced is wasted, enough to feed millions every year. In nature, waste equals food. Humanity still has work to do in learning that lesson.

In South Africa, roughly 10 million tonnes of food are wasted annually, much of it lost before reaching a plate. Globally, around 1.3 billion tonnes of food are discarded each year. This represents a staggering inefficiency across production, logistics, storage, and consumption. These losses remind us that the world produces more than enough food; the task is to use it more wisely.



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Biofuel leaders of the Global South

Brazil offers a valuable case study for the Global South. Decades of experience with sugar cane-based ethanol have shown that food and fuel can coexist successfully. While sugar demand has plateaued as consumers become more health-conscious, ethanol production has provided farmers with stable income, rural employment, and export opportunities – without compromising food availability. By integrating food and fuel markets, Brazil has demonstrated that sustainable bioenergy can strengthen, rather than weaken, food security.

Modern biofuel production has evolved far beyond the "Food vs Fuel" debate. Bioethanol refineries now use every part of the sugar cane or field corn crop. Fermentable material becomes ethanol. The remaining portion becomes Distillers Dried Grains with Solubles (DDGS), a high-protein animal feed, in the case of field corn. DDGS replaces traditional feed 1:1 in nutritional value, meaning ethanol production complements, rather than competes with, the food system. It's no longer Food vs Fuel. It's Food and Fuel.

South African innovation in biofuels

Simultaneously, technology continues to push boundaries. A South African pilot project is already converting hotel food waste into fuel-grade ethanol, proving that even discarded meals can become clean energy. Meanwhile, algae to ethanol research shows immense promise – certain strains of algae have already been isolated that produce starch rather than simple biomass.

Algae thrives on non-arable desert land under intense sunlight, and can be nourished by sewage. This approach could turn waste and sunlight into renewable energy without using food crop land, while simultaneously addressing urban wastewater challenges across the Global South.

While hydrogen and electric vehicles both aim to decarbonise transport, their real-world practicality remains limited. Hydrogen fuel cells are still far from market-ready, requiring costly storage, high-pressure systems, and entirely new distribution networks. Electric vehicles, though admirable, depend heavily on lithium, cobalt, and rare earth mining; resources that carry significant environmental and social costs, especially in developing regions.

In contrast, E85 (a fuel blend of 85% ethanol and 15% gasoline) stands out as a sustainable, balanced, immediate solution to petroleum-gasoline fossil fuel. It possesses an energy density comparable to petrol (gasoline), while hydrogen and batteries offer significantly lower energy content per kilogram. That means ethanol delivers range and performance without the weight penalties or infrastructure complications of other green technologies.

The internal combustion engine (ICE) is a mature, globally mass-produced technology which can be easily optimised for E85, combining extreme efficiency, reliability, familiarity, and affordability. Since transport remains the backbone of every economy, sustainable fuels like E85 can power growth without disruption, offering potential savings of 15–25% per kilometre compared to fossil fuels.

Meanwhile, converting ethanol into sustainable aviation fuel (SAF) remains an inefficient detour at present. It takes around 1.615 litres of ethanol to produce just 1 litre of jet fuel, which is a yield of roughly 62%. SAF production is expensive, centralised, and adds limited employment opportunities, as most airport operations already have fixed staffing structures.

In contrast, expanding E25-E85 blends for ground transport stimulates local economies by creating ongoing employment in agriculture, logistics, and retail fuel distribution. E85 benefits communities on the ground, while SAF would largely benefit the billionaire's excuse for chartering private jet trips. Adapting jet engines to run directly on bioethanol could be a better long-term engineering pathway, though such innovation may take a decade or more to materialise.

It's important to recognise that no single energy pathway is "wrong". Hydrogen, electric vehicle, and biofuel technologies all have vital roles to play in a balanced future energy mix. Diversity allows nations to pursue solutions that fit their local resources and economic realities. Even crude oil and coal will continue to serve specific purposes for decades to come. Innovation can ensure the world steadily reduces its dependence on these finite fossil resources. Diversity, not dogma, will drive true sustainability.

As an automotive enthusiast, I admire E85 for its blend of passion and purpose. It offers superior 108 RON Octane, cleaner combustion, and up to 60% lower carbon emissions, proving that performance and environmental responsibility can coexist. It's a win-win fuel that rewards engineering, thrills the driver, and respects the planet, all at once.